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The Photonic Wave Computer

A Complete Architecture for Multi-State Computation via Electromagnetic Wave Interference

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April 2026

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1. Overview

This document describes a complete photonic computing architecture in which all computation — logic gates, cascading, storage, and conditional control — is performed by electromagnetic wave interference without intermediate conversion to electronic signals. The architecture is grounded in the multiplicative structure of the integers, using Dirichlet characters as the bridge between number theory and wave physics.

The system has been developed across a series of papers and computational simulations:

Paper	Title	Core Contribution
[1]	The Logarithmic Isomorphism	Four-layer algebraic framework: number theory, Boolean logic, resistance topology, and prime coordinates are instances of a single structure

Paper	Title	Core Contribution
[2]	The Natural Operating System	Master equation $y(S) = \sum \mu(n) \cdot O_n \cdot [n \mid S] \cdot P_n(S)$ — Boolean computation via Möbius inversion
[3]	Structured Self-Information	Information spectrum $\sigma_p(n) = v_p(n) \cdot \ln(p)$ — prime coordinate decomposition of Shannon's self-information
[4]	Multi-State Wave Computation	Generalisation from binary to q-state via Dirichlet characters; polynomial cascade theorem; functional completeness
[5]	Spectral-Temporal Wave Encoding	Multi-wavelength pulse encoding; GCD/LCM as temporal overlap/union; linear scaling; knowledge investment
[—]	The Geometry of Knowing	Working notes: information is conserved when reshaped from quantity (dots) to structure (prime hypergrid) to spectrum (coordinates)

All simulation code is contained in the `wave_computer/` workspace, with the physics engine (`wave_engine.py`) shared across all steps.

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2. Theoretical Foundations

2.1 The Algebraic Foundation (Paper 1)

The logarithm $\Omega(1/n) = \ln(n)$ is a monoid isomorphism from inverse integers under multiplication to non-negative reals under addition. This is the unique continuous map with this property (Cauchy's functional equation). Under this isomorphism:

- **Multiplication ↔ addition**: $\ln(ab) = \ln(a) + \ln(b)$
- **GCD ↔ componentwise minimum** of prime coordinate vectors
- **LCM ↔ componentwise maximum**
- **Conservation**: $v(a) + v(b) = v(\text{gcd}) + v(\text{lcm})$ (componentwise)
- **Coprimality probability**: $P(\text{gcd}(a, b) = 1) = 6/\pi^2 = 1/\zeta(2)$

The prime coordinate vector $v(n) = (a_2, a_3, a_5, \ldots)$ where $n = 2^{a_2} \cdot 3^{a_3} \cdot 5^{a_5} \cdots$ provides a unique representation in a vector space spanned by the linearly independent set $\{\ln(p) : p \text{ prime}\}$.

2.2 The Master Equation (Paper 2)

The Natural Operating System computes Boolean functions via:

$$y(S) = \sum_{n \in D} \mu(n) \cdot O_n \cdot [n \mid S] \cdot P_n(S)$$

where $\mu(n)$ is the Möbius function, O_n is a database output, $[n \mid S]$ is a divisibility-based activation indicator, and $P_n(S)$ tests a condition. Inputs are encoded as sets of primes: bit A at prime 2, bit B at prime 3, giving the input integer $x = 2^A \cdot 3^B$.

The Möbius signs automatically provide the correct coefficients for inclusion-exclusion — the OR gate uses three terms with coefficients $+1, +1, -1$, arising directly from the Euler product inverse $1/\zeta(s) = \sum \mu(n)/n^s$. NOT, AND, OR, and NAND are all implementable; since NAND is functionally complete, any Boolean function can be realised.

2.3 The Information Spectrum (Paper 3)

The self-information $h(1/n) = \ln(n)$ is decomposed into independent prime dimensions:

$$\sigma_p(n) = v_p(n) \cdot \ln(p)$$

This is the unique continuous decomposition compatible with the multiplicative structure. The spectrum satisfies:

- $\sigma(ab) = \sigma(a) + \sigma(b)$ (independent sources combine additively)
- $\sigma(\gcd(a, b)) = \min(\sigma(a), \sigma(b))$ (shared information)
- $\sigma(\mathrm{lcm}(a, b)) = \max(\sigma(a), \sigma(b))$ (combined information)
- $\sigma(a) + \sigma(b) = \sigma(\gcd) + \sigma(\mathrm{lcm})$ (conservation of information)

2.4 The Geometry of Knowing (Working Notes)

The factorisation of n into a k-dimensional hypergrid is a geometric reorganisation: n dots on a line are reshaped into a grid with perpendicular axes corresponding to prime factors. The side length along the p-axis is $p^{v_p(n)}$. The logarithm converts side lengths into information coordinates, preserving perpendicularity.

The forward map (dots → grid → spectrum) is deterministic and unique. The reverse map is degenerate: the same spectrum is consistent with multiple physical orientations. This asymmetry between analysis (unique) and synthesis (degenerate) is structural — it reflects the fact that abstract structure does not determine its concrete instantiation.

The central insight for computation: the choice of representation is a *knowledge investment*. In base 10, divisibility by 2 and 5 is free (visible from the last digit) because those primes were pre-invested in the base. In the wave computer, the choice of wavelength channels is the same kind of investment.

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3. The Wave Computation Architecture (Paper 4)

3.1 The Generalised Master Equation

Binary computation via Möbius signs ($\mu(n) \in \{-1, +1\}$) generalises to q-state computation by replacing $\mu(n)$ with Dirichlet characters $\chi(n)$ of order q:

$$y(x) = \sum_{j=0}^{\varphi(m)-1} \hat{F}_j \cdot \chi_j(x \bmod m)$$

where m is a prime modulus, χ_j are the $\varphi(m)$ Dirichlet characters mod m, and $\hat{F}_j$ are complex Fourier coefficients encoding the target function. The output state is determined by the phase of the complex result:

$$\text{output} = \text{argmin}_{k \in \{0,\dots,q-1\}} \left| \arg(y(x)) - \frac{2\pi k}{q} \right|$$

3.2 Physical Implementation

Each character χ_j corresponds to a physical wave source. Source j emits with complex amplitude $\hat{F}_j \cdot \chi_j(x \bmod m)$. The superposition at the detector is the Fourier synthesis. The output state is read from the phase sector of the total amplitude.

Sources are placed on an arc of radius R = integer wavelengths centred on the detector, ensuring $e^{ikR} = 1$ (exact propagation phases). The wave engine propagates cylindrical waves with $1/\sqrt{r}$ amplitude decay.

3.3 Functional Completeness

Theorem (Paper 4, Section 5). For any $q \geq 2$ and $k \geq 1$, every function $f : \{0, \dots, q-1\}^k \to \{0, \dots, q-1\}$ is implementable.

The proof constructs the Fourier coefficients $\hat{F}_j$ via the character orthogonality relations. Every possible gate is realisable by choosing the right set of source amplitudes.

3.4 The Polynomial Cascade Theorem

Theorem (Paper 4, Section 6). Gate 1's output $z_1 = \omega_q^k$ feeds into Gate 2 via a polynomial of degree at most $q-1$ in z_1:

$$y_2 = \sum_{p=0}^{q-1} C_p \cdot z_1^p$$

Key consequences:

- **Binary ($q = 2$):** Degree 1 — purely linear. $y_2 = c_0 + c_1 z_1$. No nonlinear elements needed. Arbitrarily deep circuits cascade via linear wave operations alone.
- **Ternary ($q = 3$):** Degree 2 — requires second-harmonic generation (SHG), a standard nonlinear optical process.
- **q-state:** Degree $q - 1$ — requires $(q - 1)$-th harmonic generation.

Binary cascading being linear means the entire binary circuit is a single interference network — a mesh of linear wave paths with fixed complex weights. This is the architecture of coherent photonic processors.

3.5 Verification Results

Test	Result
Binary NOT, AND, OR, NAND	16/16 input combinations correct
All 27 single-input ternary functions	27/27 correct
All 256 single-input quaternary functions	Verified via demo mode
Ternary 2-input (sampled)	500/500 correct
All 49 named ternary gate compositions via cascade	49/49 correct
Ternary cascade NEGATE ∘ INCREMENT	3/3 correct, polynomial = z_1^2 (pure SHG)

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4. Multi-Wavelength Encoding (Paper 5)

4.1 The Scaling Problem and Its Solution

Paper 4's single-wavelength encoding requires modulus $m > \prod p_i^{q-1}$ for k inputs — exponential in k. Paper 5 resolves this by assigning each input its own wavelength channel:

Encoding	k inputs, q states	Sources
Single-wavelength (Paper 4)	$\varphi(m) \sim q^k$	Exponential
Multi-wavelength (Paper 5)	$k \cdot \varphi(m_0)$ where m_0 depends only on q	Linear

4.2 The Information Spectrum as Physical Wave Structure

The integer $n = \prod_p p^{a_p}$ is encoded as a multi-wavelength pulse packet:

- **Each prime** p → a distinct wavelength channel λ_p
- **Each exponent** a_p → pulse duration $a_p \cdot \tau$ in that channel

- **Each computational state** → phase ϕ_p from the character framework

The wave packet *is* the number. The information spectrum is a physical measurement: separate wavelength channels with a spectrograph, measure pulse durations, and the resulting vector is $\sigma(n)$.

4.3 GCD and LCM as Physical Measurements

Operation	Wave measurement	Formula
$\gcd(a,b)$	Temporal overlap (both pulses present)	$\min(v_p(a), v_p(b)) \cdot \tau$ per channel
$\mathrm{lcm}(a,b)$	Temporal union (either pulse present)	$\max(v_p(a), v_p(b)) \cdot \tau$ per channel
Conservation	Total pulse duration conserved	$T_a + T_b = T_{\mathrm{gcd}} + T_{\mathrm{lcm}}$
Coprimality	No temporal overlap in any channel	Zero overlap = coprime

No gates, no interference computation, and no arithmetic required — the lattice operations are direct physical measurements.

4.4 The Knowledge Investment Hierarchy

Level	Investment	What's free	What costs work
None (unary)	None	Nothing	Everything
Base (positional)	Factors of the base	Divisibility by base factors	Other primes
Spectral (wavelengths)	One channel per prime	All GCD/LCM, coprimality, single-input gates	Cross-channel functions
Computational (characters)	Fourier coefficients	Arbitrary q-state computation	Changing the function

The total information is conserved across all levels — only the balance between "known" and "unknown" structure shifts.

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5. Feedback, Storage, and the Ring Register

5.1 Self-Feedback Discovery

The polynomial cascade theorem, applied to a gate feeding back into itself, reveals that cyclic permutation gates (INCREMENT, DECREMENT) produce degree-1 (linear) self-cascade polynomials:

Gate	Self-cascade polynomial	Physical implementation
Binary NOT	$z \mapsto (-1) \cdot z$	Fibre loop + $\lambda/2$ phase shift
Ternary INCREMENT	$z \mapsto \omega_3 \cdot z$	Fibre loop + $\lambda/3$ phase shift
Ternary DECREMENT	$z \mapsto \omega_3^* \cdot z$	Fibre loop + $2\lambda/3$ phase shift

No nonlinear elements. No detection. No electronics. A fibre loop with a fixed phase shift creates a counter.

Binary NOT feedback is a clock oscillator (0→1→0→1…). Ternary INCREMENT feedback is a ternary counter (0→1→2→0→1→2…). The system's recurrence time for multiple channels is the LCM of the individual periods — the primorial for distinct primes.

5.2 Cavity Resonance and the Eigenmode Principle

A cavity containing a gate selects eigenmodes where the gate transformation is self-consistent. The standing wave condition is:

$$T(z) \cdot e^{ikL} = z$$

This shifts the cavity's resonant frequencies by an amount proportional to the gate phase:

$$\Delta f = \frac{\phi_{\text{gate}}}{2\pi} \cdot \text{FSR}$$

where FSR is the free spectral range. The resonant frequency *encodes* the gate function. This was verified computationally: the three ternary gates (IDENTITY, INCREMENT, DECREMENT) produce three distinctly shifted spectral signatures, with the shift exactly matching $\phi_{\text{gate}}/(2\pi) \times \text{FSR}$.

The Fabry-Perot (two-mirror) cavity exhibits an amplitude problem: off-resonance states produce near-zero signal. This is resolved by the ring cavity.

5.3 The Ring Register

A pulsed fibre ring eliminates the amplitude problem entirely. The key physics: a pulse circulating in a ring decays as r^N per round trip, *independent of the gate phase*. The gate affects only the phase, never the amplitude. This was verified computationally — all three gate types produce identical amplitude decay curves.

The ring register supports four operations, all purely optical:

Operation	Physical mechanism	Effect
SET	Inject pulse at phase $2\pi k/q$	State becomes k
HOLD	Gate = IDENTITY ($\phi = 0$)	State persists (decays slowly)
COMPUTE	Gate = any function ($\phi = 2\pi f(k)/q$)	State transforms
READ	Tap off fraction, measure phase	Non-destructive readout

5.4 Coupled Registers

Two ring registers can be coupled: Register A's stored state controls Register B's gate function. This was verified computationally:

- A = 0 → B holds its state (IDENTITY gate)
- A = 1 → B increments (INCREMENT gate)
- A = 2 → B decrements (DECREMENT gate)

The coupling is achieved by the *phase* of A's circulating pulse determining the *gate setting* of B's ring — one wave controlling another wave's transformation, without leaving the optical domain.

5.5 Interferometric Conditional Branching (JZ)

The final obstacle to a fully optical signal path was the JZ (jump-if-zero) instruction, which appeared to require phase measurement. This is resolved by the **polynomial state indicator**:

Theorem. The polynomial $f_t(z) = (1/q)\sum_{j=0}^{q-1} \omega_q^{-jt} \cdot z^j$ satisfies $f_t(\omega_q^k) = \delta_{k,t}$ (Kronecker delta) for all states k. The proof is one line: character orthogonality on $\mathbb{Z}/q\mathbb{Z}$.

The extinction is mathematically perfect ($|f_t| < 10^{-15}$ for non-target states). The polynomial has degree $q-1$ — the same as the cascade polynomial — so **JZ requires no components beyond those already needed for gate cascading.** For binary, JZ is purely linear: $f(z) = (1+z)/2$, implemented as a beam splitter combining the signal with a phase reference.

With interferometric JZ, the processor's signal path contains **no measurement at any point** between input and output. Every operation — logic, cascade, storage, branching — is wave interference.

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6. Complete Architecture Summary

6.1 Components of the Photonic Computer

Component	Physical realisation	Role	Paper
Gate	Array of phase-modulated EM sources	Computes any q-state function via interference	[4]
Cascade	Polynomial evaluation in complex amplitude	Connects gate output to gate input	[4]
Wavelength channel	Distinct optical frequency per prime	Independent computational dimensions	[5]
Pulse encoding	Duration = exponent, wavelength = prime	Represents integers as wave packets	[5]
GCD/LCM detector	Temporal overlap / union measurement	Lattice operations without computation	[5]
Register (latch)	Pulsed fibre ring with switchable gate	Stores state, transforms on command	This work
Coupled registers	One ring's phase controls another's gate	Conditional computation	This work
Counter / oscillator	Fibre ring with fixed phase shift	Sequential state cycling	This work
Clock	Beat pattern of multi-wavelength feedback	Emergent timing from prime structure	[5] + this work

6.2 Turing Completeness Status

Requirement	Status	Mechanism
Functional completeness	Proven	Fourier inversion on character groups (Paper 4, Theorem 5.1)
Gate cascading	Proven	Polynomial cascade; binary is linear (Paper 4, Theorem 6.1)
State storage	Demonstrated	Ring register with SET/HOLD/COMPUTE/READ (Step 8)
Conditional control	Demonstrated	Coupled registers: one controls another's gate (Step 8)
Loss compensation	Engineering	Erbium-doped fibre amplifiers (standard telecom technology)
Arbitrary program execution	Demonstrated	Minsky counter machine (addition) verified

The four fundamental capabilities — combinational logic, cascading, storage, and conditional control — are all demonstrated in the wave domain. The remaining gap to full Turing completeness is the control sequencing: a mechanism to read the stored state of a register and use it to determine which gate function is applied at the next step. The coupled register demonstration (Step 8) shows the *mechanism* for this — one register controlling another's gate — but a full Turing-complete controller would require a more elaborate arrangement.

6.3 Physical Encoding Dimensions

The architecture uses six physical properties of electromagnetic waves:

Property	Encodes	Role
Wavelength (λ)	Which prime / which channel	Input/output identity
Phase (ϕ)	Computational state (q-th root of unity)	Gate logic via interference
Pulse duration (T)	Prime exponent ($v_p(n)$)	Integer encoding
Time (t)	Synchronisation, overlap windows	GCD/LCM, temporal logic
Distance (L)	Propagation phase, loop transit time	Source geometry, feedback period
Intensity ($\lvert E\rvert^2$)	Signal strength	Threshold detection, bistability (future)

7. Simulation Workspace

7.1 Architecture

All simulation code is in `wave_computer/`. The physics engine `wave_engine.py` owns all wave propagation; individual step scripts import from it and never reimplement the physics.

```
wave_computer/
├── wave_engine.py          # Core: WaveSource, compute_field, measure_at_point, detect_pl
├── step1_single_source.py      # Single EM point source
├── step2_binary_interference.py # Two-source interference, binary NOT
├── step3_binary_gates.py     # Full binary suite: NOT, AND, OR, NAND (6 sources, m=7)
├── step4_ternary_gates.py    # Ternary gates, 27/27 verified (6 sources, m=7)
├── step5_ternary_cascade.py   # Polynomial cascade: NEGATE ∘ INCREMENT = z₁² (SHG)
├── step6_interactive.py      # Interactive explorer: q=2,3,4, switchable gates
├── step7_cavity_eigenmode.py  # Cavity resonance: gate shifts eigenfrequency
├── step8_ring_register.py    # Ring register: SET, HOLD, COMPUTE, coupled registers
├── build_executable.py         # PyInstaller packaging
└── outputs/                    # Generated images and animations
```

7.2 Key Design Principle

All computation happens in the wave physics. The simulation propagates electromagnetic waves from point sources, computes interference at every spatial point, and measures the result at a detector. No shortcut arithmetic is used — gate outputs emerge from physical wave superposition. The wave engine does not know what gate is being computed; it just propagates waves.

7.3 Running the Simulation

```
python3 -m venv venv && source venv/bin/activate
pip install numpy matplotlib pillow
python step1_single_source.py       # Verify wave rendering
python step3_binary_gates.py        # Binary gate suite
python step4_ternary_gates.py       # Ternary gates (all 27 functions)
python step5_ternary_cascade.py     # Polynomial cascade (49/49 compositions)
python step6_interactive.py         # Interactive explorer (GUI)
python step6_interactive.py --demo  # Demo mode (no GUI needed)
python step7_cavity_eigenmode.py    # Cavity eigenmode computation
python step8_ring_register.py       # Ring register and coupled registers
```

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8. Open Problems

8.1 Toward Full Turing Completeness

The coupled register mechanism (Section 5.4) provides conditional control — one register's state determines another's gate function. A Turing-complete controller would chain multiple coupled registers into a finite state machine, where the current instruction (stored in a control register) determines the operation applied to data registers, and the result updates the control register. The mechanism exists; the detailed design of the controller architecture is an open problem.

8.2 Loss Compensation Without Detection

The circulating pulse in a ring register decays by a factor r per round trip ($r \approx 0.98$ in our simulations). Over many cycles, this degrades the signal. Erbium-doped fibre amplifiers (EDFAs) amplify the optical signal without detecting or re-encoding it — they preserve the phase information while restoring the amplitude. Integrating an EDFA into the ring loop is standard telecommunications engineering, but the noise figure of the amplifier introduces phase noise that accumulates over many cycles. The noise tolerance analysis (Step 5) shows that the ternary system tolerates $\sigma \approx 0.05$ rad/pass for 200+ error-free iterations.

8.3 Nonlinear Cross-Channel Computation

The linear scaling of Paper 5 comes at the cost of requiring nonlinear frequency mixing for multi-input functions. Sum-frequency generation, difference-frequency generation, and four-wave mixing are established technologies, but their integration into the character framework — designing specific nonlinear interactions that implement the correct gate functions across wavelength channels — is theoretically undeveloped.

8.4 The Eigenmode Computer

The cavity eigenmode principle (Step 7) suggests a fundamentally different computational paradigm: instead of sequential gate application, the system solves for its own equilibrium. The cavity settles into the eigenmode determined by the wave equation, and the eigenmode *is* the answer. Developing this into a general-purpose computational model — where the cavity's resonant structure encodes an arbitrary program — is an open and potentially deep problem.

8.5 Connection to the Zeta Function

The Euler product $L(s,\chi) = \prod_p (1 - \chi(p)p^{-s})^{-1}$ is a native program in the wave architecture: each prime contributes a factor determined by its character value. The L-function partial sums are physically realisable as wave superpositions. Whether the analytic properties of L-functions (zero distribution, functional equation) are detectable within the wave architecture is an open question that connects the computational framework to deep problems in analytic number theory.

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9. The Five-Layer Correspondence

Paper 5's spectral-temporal encoding adds a fifth physical layer to the four-layer framework of Paper 1:

Layer	Integer n	GCD	LCM	Complement	Source
Natural language	The number n	Shared factors	Combined factors	Non-factors	[1]
Inverse integer	$1/n$	$1/\gcd(a,b)$	$1/\operatorname{lcm}(a,b)$	$(a-1)/a$	[1]
Resistance	$\Omega_n = \ln(n)$	Ω_{gcd}	$\Omega_a + \Omega_b - \Omega_{\text{gcd}}$	Phase interference	[1]
Prime coordinates	$(v_2, v_3, v_5, \ldots)$	$\min(v_a, v_b)$	$\max(v_a, v_b)$	Not closed	[1]
Wave packet	**Multi-λ pulse**	**Temporal overlap**	**Temporal union**	**Destructive interference**	[5]

The fifth layer is the first that is *directly physical*: the operations are not mathematical abstractions but measurable properties of electromagnetic waves in spacetime.

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10. References

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Appendix A: Master Notation Table

Symbol	Meaning	First appears
$\mu(n)$	Möbius function	[2]
χ_j	j-th Dirichlet character mod m	[4]
$\hat{F}_j$	Fourier coefficient / database entry for character j	[4]
ω_q	Primitive q-th root of unity: $e^{2\pi i/q}$	[4]
m	Encoding modulus (prime)	[4]
g	Generator of $(\mathbb{Z}/m\mathbb{Z})^*$	[4]
$\varphi(m)$	Euler totient: number of characters mod m	[4]
$v_p(n)$	Exponent of p in factorisation of n	[1]
$\sigma_p(n)$	Information spectrum component: $v_p(n) \cdot \ln(p)$	[3]
Ω_n	Informational resistance: $\ln(n)$	[1]
$\mathcal{P}$	Set of invested primes (assigned wavelength channels)	[5]
λ_p	Carrier wavelength assigned to prime p	[5]
τ	Base time unit for pulse duration	[5]
r	Round-trip survival factor (ring cavity)	Step 8
$L(s,\chi)$	Dirichlet L-function	[4]
$\zeta(s)$	Riemann zeta function	[1]
η	Encoding efficiency: $q/\varphi(m)$	[4]
R	Source arc radius (integer wavelengths)	Step 3
FSR	Free spectral range of cavity	Step 7